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CSC 211 — Past Questions & Materials
by Renance

Course: CSC 211
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — CSC211- 2017-2018-2019

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCES
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
2017/2018 FIRST SEMESTER EXAMINATION

CSC 211 – Computer Programming I

INSTRUCTIONS

- (i) Answer Four questions in all; Question 1 and Question 2 are compulsory.
- (ii) Duration of exams: 2½ Hrs.
- (iii) Do not mix up questions. Start a new question on a fresh page.
- (iv) All extra sheets used should be signed by the Invigilator

Question one (20marks)

- (a) List and explain using diagrams, the basic building blocks of programming.
- (b) Write a complete C program to convert temperatures from Celsius to Fahrenheit assume that $F = 32 + \left(\frac{9}{5}\right)C$.
- (c) List any four format specifiers defined in C and state the data type they represent.
- (d) Determine which of the following are valid identifiers? If the alleged identifier is not valid, give a short explanation.
 - (i) A+b (ii) h2ls (iii) real (iv) down_under (v) C fun (f) chaar

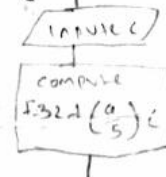
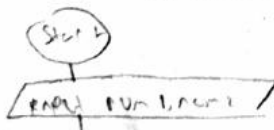
Question two (20marks)

- (a) Outline the basic structure of a C program
- (b) Briefly list and explain the guidelines or steps necessary to write a good program
- (c) Write a C program to create a menu driven calculator capable of performing the add, subtract, multiply, and divide arithmetic operations. The calculator program should accept two integer input numbers as well as an operator from user. The program should perform operation according to the operator entered. The input format is given as: num1, opr, num2

Question three (15marks)

- (a) Write a C program that accepts the year of a person's birth which is between 1900 and 2009. If an invalid entry is entered, the program should display the message, "sorry, that is not a valid year." and read year again. If the correct value is not provided after 3 attempts, the program should display the message "sorry, please you are having problems." and terminate. Once a valid year is read, the program should display the year and a message "you were born in the 20th century" Or "you were born in the 21st century". 20th century counts from 1900 to 1999 while 21st century counts from 2000 – 2099.
- (b) Write down the output of the following program?

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int alpha, num;
    for(alpha='A'; alpha<='E'; alpha=alpha+1)
    {
        for(num=1; num<=5; num=num+1)
        {
            printf("%c%d\t", alpha, num);
        }
        printf("\n");
    }
    return(0);
}
```



Handwritten note: #include <stdio.h>

Question four (15marks)

- (a) Write a C program that calculates the average age of students in your class. The program should continue to accept age of students in your class until zero (0) is entered. A negative age is considered invalid while a positive age is considered valid. Hence the program should only calculate average of the valid ages entered. The following information should be displayed finally: Total ages entered (excluding 0), Number of Valid Ages, Number of invalid ages, and Average of valid ages.
- (b) Draw a flowchart to represent the program written in **1(a)**, above.

Question five (15 marks)

Suppose that you have a task to compute area and perimeter of a circle. Assume that the $Perimeter = 2\pi r$ and that the $Area = \pi r^2$.

- Write an algorithm
- Draw a flowchart
- Write a good program in C that calculates the Area and Perimeter of the Effurun roundabout.

Your program should accept the value of r from the user and then display the Area and the Perimeter.





COS/2008/2017
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2018/2019

Course Code/Title: CSC 211 – Computer Programming I

Instructions: Attempt Question 1 and any other 3 questions. **Time Allowed:** 2 1/2 Hours.

The narration below applies to parts of Question one to Question four.

It is no longer news that some Road transport companies have online seat reservation systems. A C program can be used to simulate the features of a typical seat reservation system that offers five (5) different main menu options when the program is executed as follows:

- 1 – View available seats
- 2 – Reserve a seat
- 3 – View my reservation
- 4 – View all reservations
- 0 – Quit

If any of 1 to 4 is chosen, the appropriate task is executed and the main menu displays again.

Here is the beginning section of the program before the main function.

```
int numberOfReservations = 0;
void displayMenu(void); // This function displays the main menu

// This function displays the available seats. Available seats refer to values in the array seat that are not zero
void viewAvailableSeats(int seats[]);

/* This function reserves a seat. When a seat is reserved, three things happen.
 * First, first name, last name, password and seat number of passenger is added to array reservations.
 * Each information is at most 19 characters long.
 * Secondly, the variable numberOfReservations is incremented and thirdly, the seat number in the array seat is reset to zero.
 */
void reserveSeat(int seat[], char reservations[][4][20]);

/* This function displays a user's reservation detail.
 * To determine his reservation, the user is prompted for his password which is then compared with those in the array reservations.
 */
void viewMyReservation(char reservations[][4][20]);
void viewAllReservations(char reservations[][4][20]);
void quit(char reservations[][4][20]);
```

Question one (22 Marks)

- a. Complete the incomplete function *main* (given below) for the program (16 Marks)

```
int main(void){
    int seats[10] = {1,2,3,4,5,6,7,8,9,10}; // seat numbers from 1
    char reservations[10][4][20] = {"\0"};
    // Some code ought to be here...
    return 0;
}
```

- b. What is a sentinel value?(illustrate) (6 Marks)

Question two (16 Marks)

- a. Write the code for the function *viewAvailableSeats* (8Marks)
 b. What is a garbage value? (illustrate) (4 Marks)
 c. What is printed by the following C code segment? (4 Marks)

```
int x = 17;
int y = 20;
if (x <= 20 && y > x){
    if (x != 17)
        x = y + x/x;
    else
        x = y - x;
}
if (x < 10 || y < 20){
    if (x != 17)
        x = y + x/y;
    else
        x = x + y;
}
puts(x);
```

Question three (16 Marks)

- a. Write the code for the function *quit* (12 Marks)

The quit function should work as follows:

All the information in the array *reservations* should be saved in file called *reservation.dat*, one reservation per line. Afterwards, a message should be displayed that says: "All reservations have been saved" and then the program terminates.

- b. What is a counter variable? (2 Marks)
 c. What is a total variable? (2 Marks)

Question four (16 Marks)

- Mention three (3) C libraries that need to be included in the program and explain why? (6 Marks)
- Why does the main function's prototype start with the keyword **int** but those of the other functions in the program start with the keyword **void** (4 Marks)
- Explain why the array reservations is declared in the main function as

`char reservations[10][4][20]` (6 Marks)

(Hint: what's the significance of the numbers 10, 4 and 20 in the declaration?)

Question five (16 Marks)

- If a die is tossed, there are six possible outcomes. Write a program that toss a die six million times and display the number of times each outcome occurs. (8 Marks)
- Illustrate how a division-by-zero problem can be avoided (4 Marks)
- How many lines of output are printed by the following nested loop? (4 Marks)

```
int i = 1;
while (i <= 5){
    int j = 1;
    while (j <= i){
        puts(j);
        j++;
    }
    i++;
}
```

Question six (16 Marks)

- What value is assigned to ticket by the nested if statements on the left and right when speed is 55? Which construct is logically correct? (6 Marks)

<pre>If (speed > 35) ticket = 10.0 else if (speed > 50) ticket = 20.0 else if (speed > 70) ticket = 30.0</pre>	<pre>If (speed > 70) ticket = 30.0 else if (speed > 50) ticket = 20.0 else if (speed > 35) ticket = 10.0</pre>
--	--

- Describe integer division problem and illustrate how it can be avoided? (illustrate) (6 Marks)
- What is displayed by the following code? (4 Marks)

```
void changeMe(int n, int m){
    n = 100;
    m = 100;
}
int main(void){
    int number = 99;
    int myArray[] = {99,98,97};
    changeMe(number, myArray[0]);
    printf("%d %d", number, myArray[0]);
    return 0;
}
```

100, 100

Paper 2 of 3 — CSC 211/COS 201 Test Apst.Sam

Level 200 • First Semester • PastQuestion

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Federal University of Petroleum Resources, Effurun
1st Semester 2017/2018 Academic Session
CSC 211: Computer Programming I
Assessment

✓BQ

5
7
2
3
3:

1. Given a number X,
Draw a flowchart to check whether the number is even or odd.
Write a corresponding C program to check whether X is even or odd. 5marks
2. The solution of a quadratic equation $ax^2 + bx + c = 0$ is given by X1 and X2 where:

$$X1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$X2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Write a program to read in the coefficients a, b and c, and compute and display the roots X1 and X2. You can assume that d, given by , is always non-negative. 7marks

3. Decision control structure is responsible for handling situations in programming where alternative execution of instructions is required. List 4 examples of decision control statements. 2marks
4. Variable declaration is an important factor in programming ,given a set of variables:

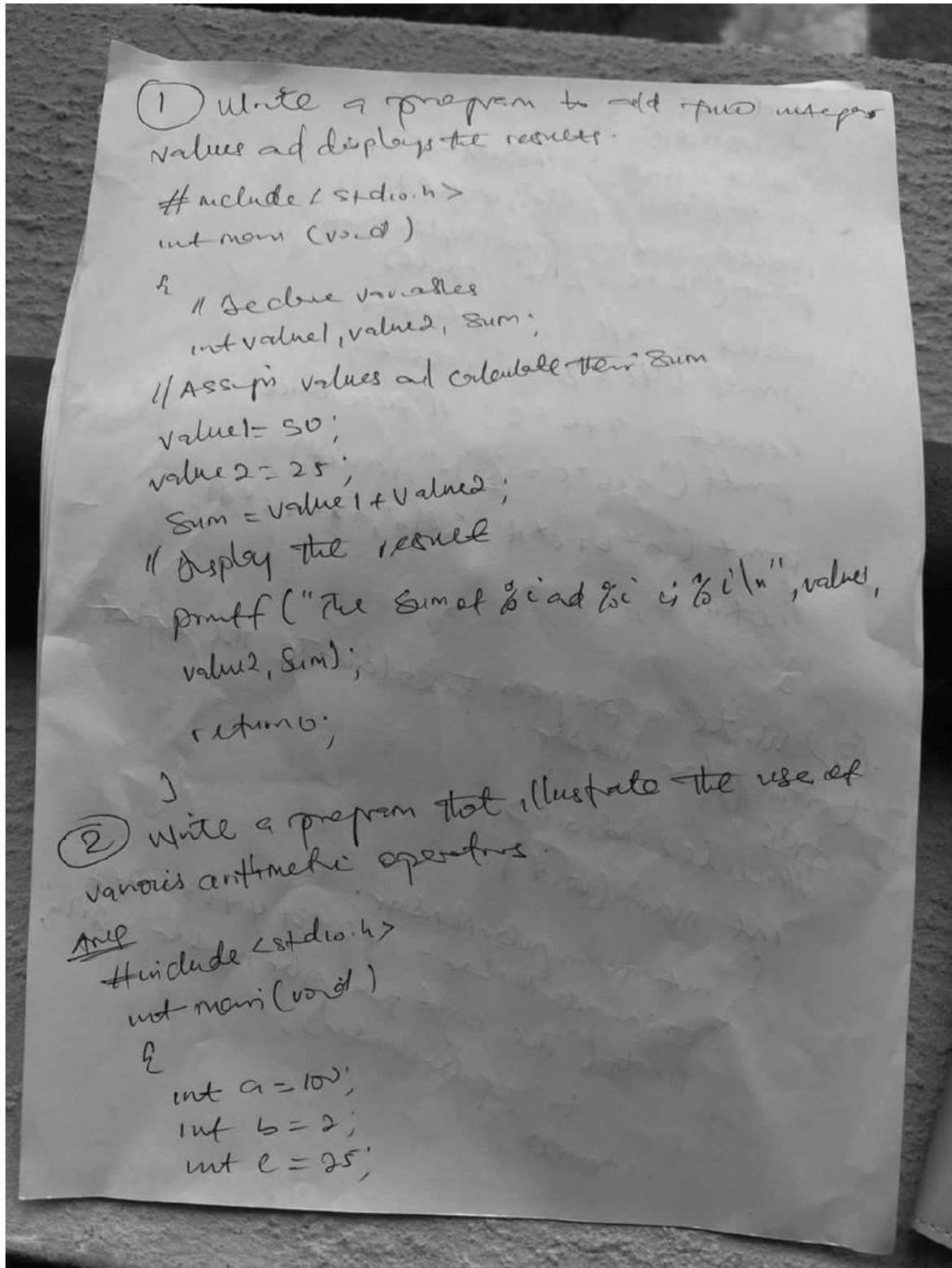
Name = Ese , salary =89.56, Num_of_Students . Write a code segment to declare them as you would in a program. 3marks
5. Mention and **explain** the basic building blocks of programming. 3marks



Paper 3 of 3 — CSC 211/ COS 201 Q&A

Level 200 • First Semester • PastQuestion

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```

int d = 4;
int result;
result = a - b; // subtraction
printf("a - b = %i\n", result);
result = b * c; // multiplication
printf("b * c = %i\n", result);
result = a / c; // division
printf("a / c = %i\n", result);
result = a + b * c; // precedence
printf("a + b * c = %i\n", result);
printf("a * b + c * d = %i\n", a * b + c * d);
return 0;
}

```

(3) Write a program to calculate the right triangular number.

Ans

#include <stdio.h>

int main(void)

{ int triangularNumber;

triangularNumber = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8

printf("The right triangular number is %i\n", triangularNumber);

return 0;

y.

```

triangularNumber = 0;
for (n = 1; n <= 10; ++n) {
    triangularNumber += n;
    printf ("%i\t\t\t\t\t%i\n", n, triangularNumber);
}
return 0;

```

Output

n	Sum from 1 to n
1	1
2	3
3	6
4	10
5	15

⑥ Using a while statement, write a program that merely counts from 1 to 5.

Ans

// program to introduce the while statement

```

#include <stdio.h>
int main (void)
{
    int count = 1;
    while (count <= 5) {
        printf ("%i\n", count);
        ++count;
    }
}

```

④ Using a for statement, write a program to calculate the 200th triangular number.

Ans

```
#include <stdio.h>
```

```
int main(void)
```

```
{
    int n, triangularNumber;
    triangularNumber = 0;
```

```
    for (n = 1; n <= 200; n = n + 1)
```

```
        triangularNumber = triangularNumber + n;
```

```
    printf("The 200th triangular number is %i\n",
           triangularNumber);
```

```
    return 0;
```

```
}
```

⑤ Using a for statement, write a program to generate a table of triangular numbers.

Ans

```
#include <stdio.h>
```

```
int main(void)
```

```
{
    int n, triangularNumber;
```

```
    printf("TABLE OF TRIANGULAR NUMBERS\n\n");
```

```
    printf("n    Sum from 1 to n\n\n");
```

```
    printf("----\n\n");
```

}
 return 0;
 }
 ② Write a program to find the greatest common divisor of two non-negative integer value.

Ans

```

#include <stdio.h>
int main(void)
{
    int u, v, temp;
    printf("Please type in two nonnegative integers.\n");
    scanf("%i%i", &u, &v);
    while (v != 0) {
        temp = u % v;
        u = v;
        v = temp;
    }
    printf("Their greatest common divisor is %i\n", u);
    return 0;
}
  
```

⑧ write a program to reverse the digits of a number using a while statement.

Ans

```
#include <stdio.h>
int main()
{
    int number, right_digit;
    printf("Enter your number: \n");
    scanf("%i", &number);
    while (number != 0)
    {
        right_digit = number % 10;
        printf("%i", right_digit);
        number = number / 10;
    }
    printf("\n");
    return 0;
}
```

⑨ Repeat the above program using the do statement

```
#include <stdio.h>
int main()
{
    int number, right_digit;
```

```
printf("Enter your number: \n");
scanf("%i", &number);
```

```
do {
```

```
    right_digit = number % 10;
```

```
    printf("%i", right_digit);
```

```
    number = number / 10;
```

```
} while (number != 0);
```

```
printf("\n");
```

```
return 0;
```

```
}
```

(10) Write a program to calculate the absolute value of an integer

Ans

```
int main(void)
```

```
{
```

```
    int number;
```

```
    printf("Type in your number: ");
```

```
    scanf("%i", &number);
```

```
    if (number < 0)
```

```
        number = -number;
```

```
    printf("The absolute value is %i\n",
```

```
        number);
```

```
    return 0;
```

(Q) Write a program to calculate the average of a set of grades and count the number of failing test grades.

Ans

```
#include <stdio.h>
int main(void)
{
    int number_of_grades, i, grade;
    int gradeTotal = 0;
    int failureCount = 0;
    float average;
    printf("How many grades will you be entering?");
    scanf("%i", &number_of_grades);
    for(i = 1; i <= number_of_grades; ++i)
    {
        printf("Enter grade # %i: ", i);
        scanf("%i", &grade);
        gradeTotal = gradeTotal + grade;
        if(grade < 65)
            ++failureCount;
    }
    average = (float) gradeTotal / number_of_grades;
    printf("\n Grade average = %.2f\n", average);
    printf("Number of failures = %i\n", failureCount);
}
return 0;
```

Sol: How many grade will you be entering??

Enter grade #1: 93

if (precision mod 100 < 10)

Create array =

Number of failures =

(12) Program to determine if a number is even or odd.

Sol

#include <stdio.h>

int main(void)

{

int number_to_test, remainder;

printf("Enter your number to be tested:");

scanf("%i", &number_to_test);

remainder = number_to_test % 2;

if (remainder == 0)

printf("The number is even.\n");

if (remainder != 0)

printf("The number is odd.\n");

return 0;

}

if (remainder == 0)

```
(3) A program to determine if a year is a leap year.  
#include <stdio.h>  
int main(void)  
{  
    int year, rem_4, rem_100, rem_400;  
    printf("Enter the year to be tested: ");  
    scanf("%i", &year);  
    rem_4 = year % 4;  
    rem_100 = year % 100;  
    rem_400 = year % 400;  
    if ((rem_4 == 0 && rem_100 != 0) || rem_400 == 0)  
        printf("It's a leap year. \n");  
    else  
        printf("Nope, it's not a leap year. \n");  
    return 0;  
}
```